Factory-to-Customer Shipping Route Efficiency at Nassau Candy Distributor: An Exploratory Analysis of 10,191 Shipments (2024–2025)

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Abstract

Nassau Candy Distributor ships products from five factories to customers across the United States and Canada, yet it has no route-level view of which lanes are fast, slow or delay-prone. This study analyses 10,191 order lines placed between January 2024 and December 2025. The supplied ship dates carried a systematic offset of 904–1,642 days, which I diagnosed as three date-offset clusters and corrected using Same Day shipments as a zero anchor. Calibrated lead time averages 3.96 days (SD 1.76), and 6.4% of shipments exceed a six-day threshold. Ship mode explains most of the variation in lead time (rank-based effect size $\varepsilon^2 = 0.59$); customer region and factory explain about 0.1% each. The best-to-worst gap among 58 ranked factory-to-state routes (1.38 days) is no larger than a permutation test produces from random labels ($p = 0.79$), so route league tables and apparent geographic bottlenecks should not be treated as structural. The most defensible levers are the Standard Class tail and data capture: recording true delivery dates and carrier cost.

Keywords: shipping lead time, route efficiency, logistics analytics, ship mode, data quality, exploratory data analysis

1. Introduction

Shipping performance directly affects customer satisfaction, operating cost and the ability of a distributor to scale. Nassau Candy holds rich order and shipment records, but logistics decisions are made without route-level efficiency intelligence, so optimisation is reactive. A route is defined here as *factory location → customer state* (or customer region).

The study addresses four questions:

- **RQ1.** Which factory-to-customer routes are consistently fast, and which are slow or delay-prone?
- **RQ2.** How does shipping performance vary by region, state and ship mode?
- **RQ3.** Where do geographic bottlenecks (high volume combined with poor performance) exist?
- **RQ4.** Are observed differences large enough, relative to normal variation, to justify operational action?

2. Data

The dataset contains 10,194 order lines with 18 fields: order and ship dates, ship mode, customer geography (country, city, state/province, postal code, region), product and division, and sales, units, cost and gross profit. Orders span 2 January 2024 to 31 December 2025 and cover 15 products, four ship modes, four customer regions and 59 states/provinces (200 lines are Canadian). Each product is made at exactly one of five factories (Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, The Other Factory), so a product's factory is fixed by the supplied product-factory table. After cleaning, 10,191 lines remain, with total sales of $141,737 and gross profit of $93,410 (margin 65.9%). Average order-line value is small (about $13.91).

3. Methodology

3.1 Cleaning and validation

Dates were parsed as DD-MM-YYYY and numeric fields coerced and checked (sales and units greater than zero). Text fields were trimmed and standardised, product names were normalised to match the product-factory table, and US postal codes were restored to five digits. Gross profit equals sales minus cost in every row (no violations). No dates were missing or unparseable. 313

lines are identical in every field except Row ID; they may be genuine repeat line items, so they were flagged and retained.

3.2 A systematic date-offset anomaly and its correction

Raw ship-minus-order gaps ranged from 904 to 1,642 days, which is impossible for candy shipments. The values fall in three tight clusters whose baselines are 904, 1,269 and 1,635 days (Figure 1), consistent with an export artefact that shifted ship dates. Each cluster was identified by splitting sorted gaps wherever consecutive values jump by more than 30 days. Within each cluster the baseline was set to the modal raw gap of Same Day shipments, which ship on the order date and therefore have a true lead time of about zero; the mode, not the minimum, is used so that stray records cannot shift a whole cluster. The baseline was subtracted from every record in the cluster. Three records became negative after calibration and were removed as invalid. After calibration the ship modes separate cleanly and plausibly (Same Day ≈ 0 days, First Class ≈ 2, Second Class ≈ 3, Standard Class ≈ 5).

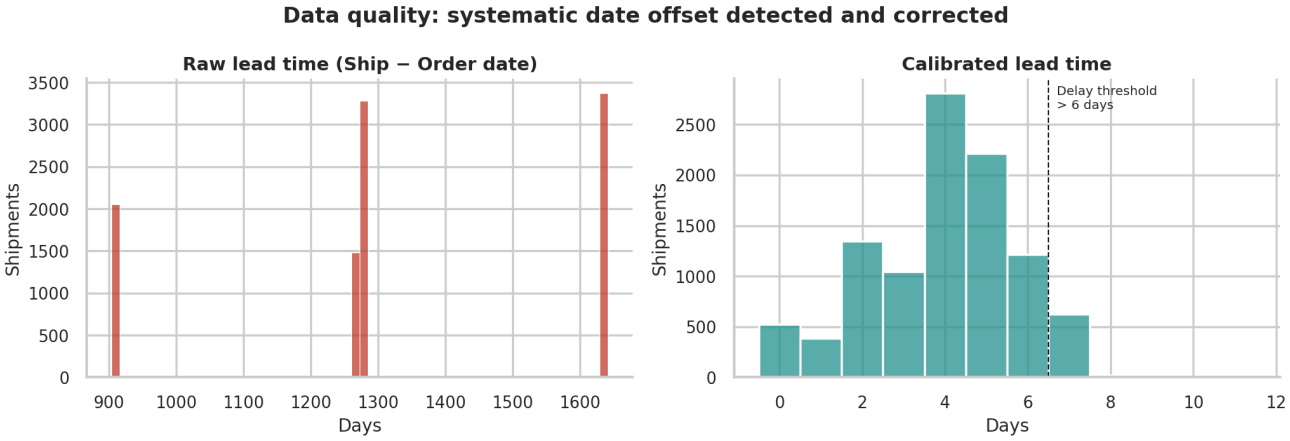


Figure 1. Raw ship-minus-order gap (left) versus calibrated lead time (right) with the six-day delay threshold.

Caveat. The correction rests on the assumption that Same Day lead time is zero. Absolute levels therefore depend on that assumption; comparisons between modes, routes and regions within the corrected data do not depend on it as long as each cluster is shifted consistently.

3.3 Features, KPIs and benchmarking

Table 1. Key performance indicators.

KPI	Definition
Shipping lead time	Ship date minus order date, calibrated (days)
Average lead time	Mean lead time per route, state, region or mode
Route volume	Number of shipped order lines on the route
Delay frequency	Share of shipments with lead time above the threshold (default 6 days; sensitivity in Section 5.5)
Route efficiency score	Min-max normalised average lead time across ranked routes, 0 (slowest) to 100 (fastest)
Mix-adjusted lead time	Lead time minus the mean for its ship mode, so routes are not credited for using faster modes

State routes were ranked only when they carried at least 30 shipments, which leaves 58 of the possible factory-to-state routes. A state is flagged as a bottleneck when its volume is at or above the median state volume and its average lead time exceeds the national average.

3.4 Statistical testing

Lead times are discrete and right-skewed, so group differences were tested with the Kruskal-Wallis test and summarised by the rank-based effect size $\varepsilon^2 = (H - k + 1)/(n - k)$. Because route league tables select extremes from many noisy averages, I also ran a permutation test: lead times were shuffled within each ship mode (keeping route sizes fixed) 2,000 times, and the best-to-worst route gap was recomputed each time to see how large a gap chance alone produces.

4. Exploratory Data Analysis

Volume is concentrated. Lot's O' Nuts ships 55.8% of lines and Wicked Choccy's 40.7%; the other three factories together ship 3.5%. Chocolate accounts for 96.6% of lines. As a result, routes from Sugar Shack, The Other Factory and Secret Factory rest on very small samples.

Standard Class dominates. Standard Class carries 60.1% of shipments, Second Class 19.4%, First Class 15.2% and Same Day 5.3%.

Geography. Average lead time by customer region ranges only from 3.92 days (Atlantic) to 4.05 days (Interior). Calibrated lead time averaged 4.07 days for 2024 orders and 3.89 days for 2025 orders, a small difference that is descriptive only.

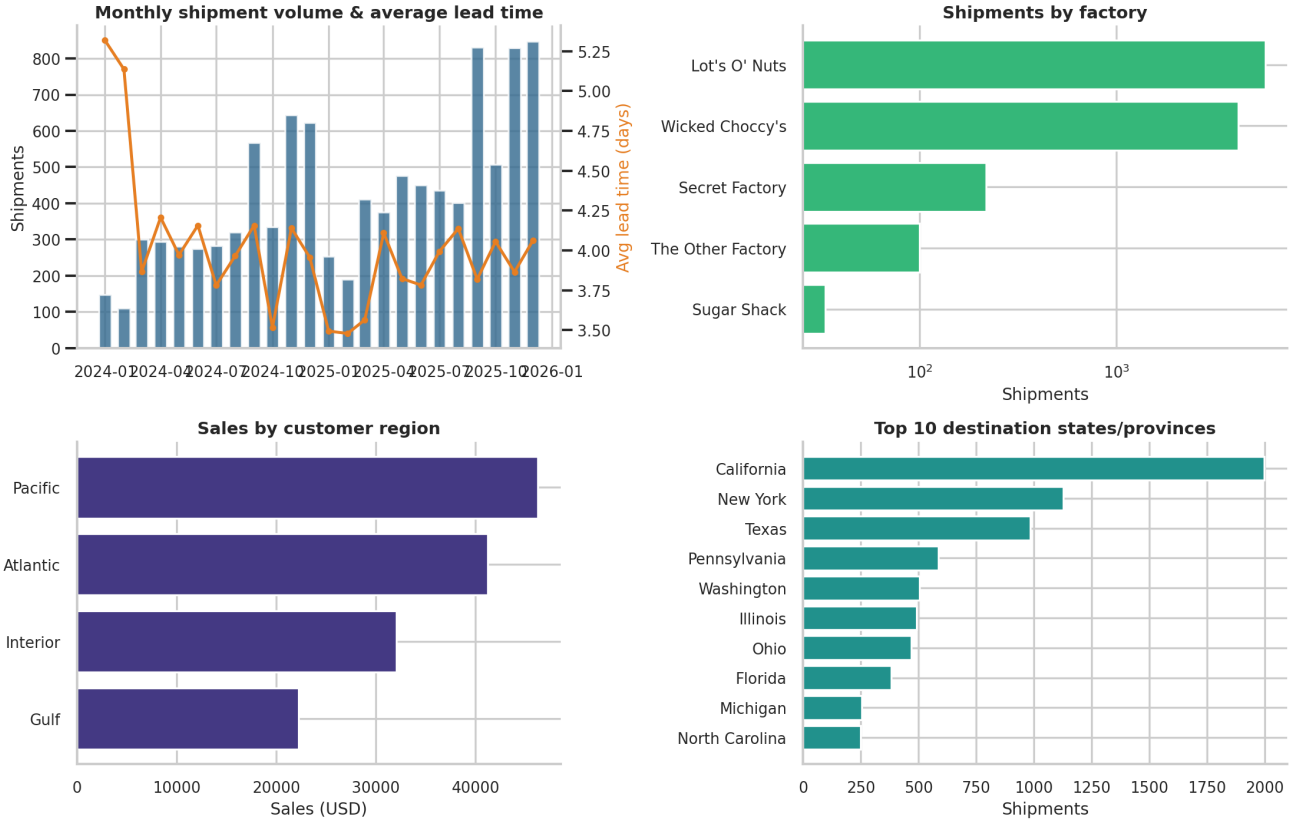


Figure 2. Monthly volume and lead time, shipments by factory (log scale), sales by region and top destinations.

5. Results

5.1 Ship mode

Table 2. Performance by ship mode.

Ship mode	Shipments	Avg lead time (days)	SD (days)	Delayed (>6 days)	Avg gross margin
Same Day	544	0.05	0.21	0.0%	66.5%
First Class	1,548	2.18	0.79	0.0%	66.5%
Second Class	1,979	3.24	1.21	0.0%	66.5%
Standard Class	6,120	5.00	1.04	10.6%	66.5%

Lead time rises monotonically from Same Day to Standard Class, and the modes are clearly separated. All delayed shipments (648) are Standard Class, 10.6% of that mode. Gross margin is essentially identical across modes (66.5% to 66.6%), so within these data faster service does not show up as lower margin. This is a descriptive cost-time view only, because actual freight cost is not recorded.

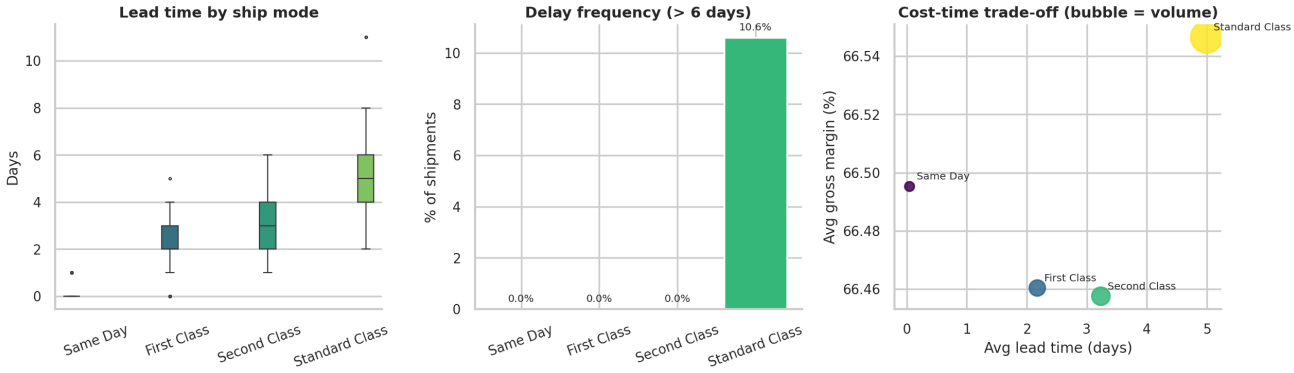


Figure 3. Lead time distribution, delay frequency and margin versus lead time by ship mode.

5.2 Route benchmarking

Table 3. Ten fastest factory-to-state routes (minimum 30 shipments).

Route	Shipments	Avg lead time	SD	Delayed
Lot's O' Nuts to Rhode Island	30	3.20	1.88	13.3%
Lot's O' Nuts to Ohio	262	3.47	1.98	5.0%
Wicked Choccy's to Ohio	189	3.47	1.87	4.8%
Lot's O' Nuts to Connecticut	47	3.49	1.88	0.0%
Wicked Choccy's to Colorado	70	3.60	1.39	0.0%
Secret Factory to California	31	3.61	1.73	3.2%
Wicked Choccy's to Connecticut	31	3.61	2.26	9.7%
Wicked Choccy's to Ontario	30	3.63	1.07	0.0%
Lot's O' Nuts to Massachusetts	76	3.64	1.61	5.3%
Lot's O' Nuts to Colorado	103	3.76	1.80	7.8%

Table 4. Ten slowest factory-to-state routes (minimum 30 shipments).

Route	Shipments	Avg lead time	SD	Delayed
Lot's O' Nuts to Minnesota	43	4.58	1.35	14.0%
Lot's O' Nuts to Oklahoma	41	4.56	2.00	9.8%
Wicked Choccy's to Indiana	54	4.56	1.48	7.4%
Wicked Choccy's to Delaware	49	4.55	1.51	4.1%
Lot's O' Nuts to New Jersey	65	4.51	1.56	10.8%
Wicked Choccy's to Tennessee	67	4.48	1.34	7.5%
Lot's O' Nuts to Alabama	34	4.47	1.54	14.7%
Wicked Choccy's to Minnesota	45	4.47	2.03	17.8%
Wicked Choccy's to Oregon	43	4.44	1.62	4.7%
Lot's O' Nuts to Oregon	77	4.40	1.58	3.9%

The fastest ranked route is Lot's O' Nuts to Rhode Island (3.20 days, n = 30) and the slowest is Lot's O' Nuts to Minnesota (4.58 days, n = 43). The full gap is 1.38 days, against a per-shipment SD of about 1.8 days. **This gap is not distinguishable from chance:** across 2,000 permutations the expected gap is 1.55 days (95th percentile 1.93), and the observed gap is not larger than that (p = 0.79). The leaderboards are therefore a ranking of noise as much as of lanes, and routes with 30–60 shipments in particular should not be acted on individually.

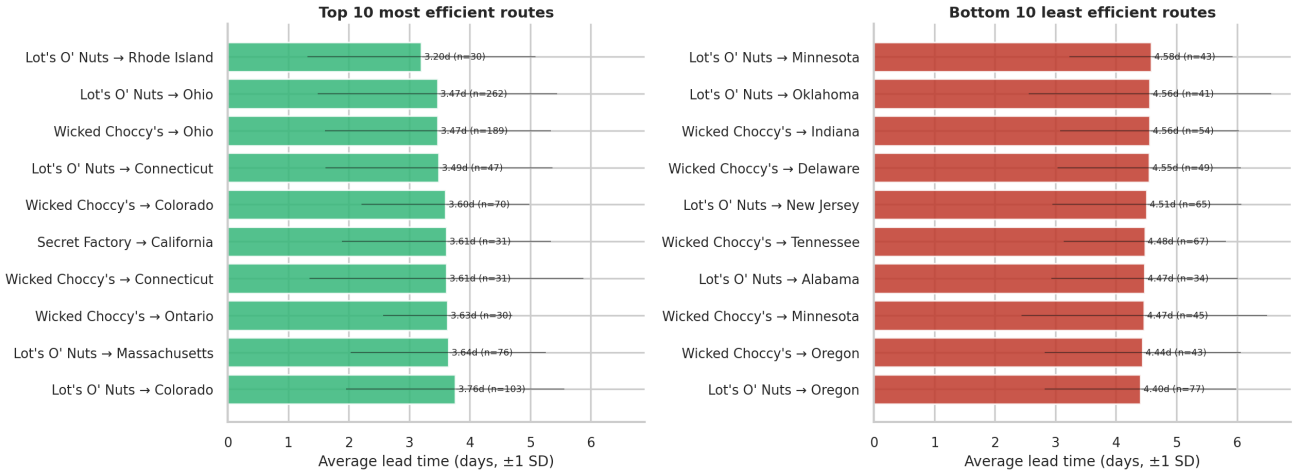


Figure 4. Top 10 and bottom 10 routes by average lead time (error bars ±1 SD).

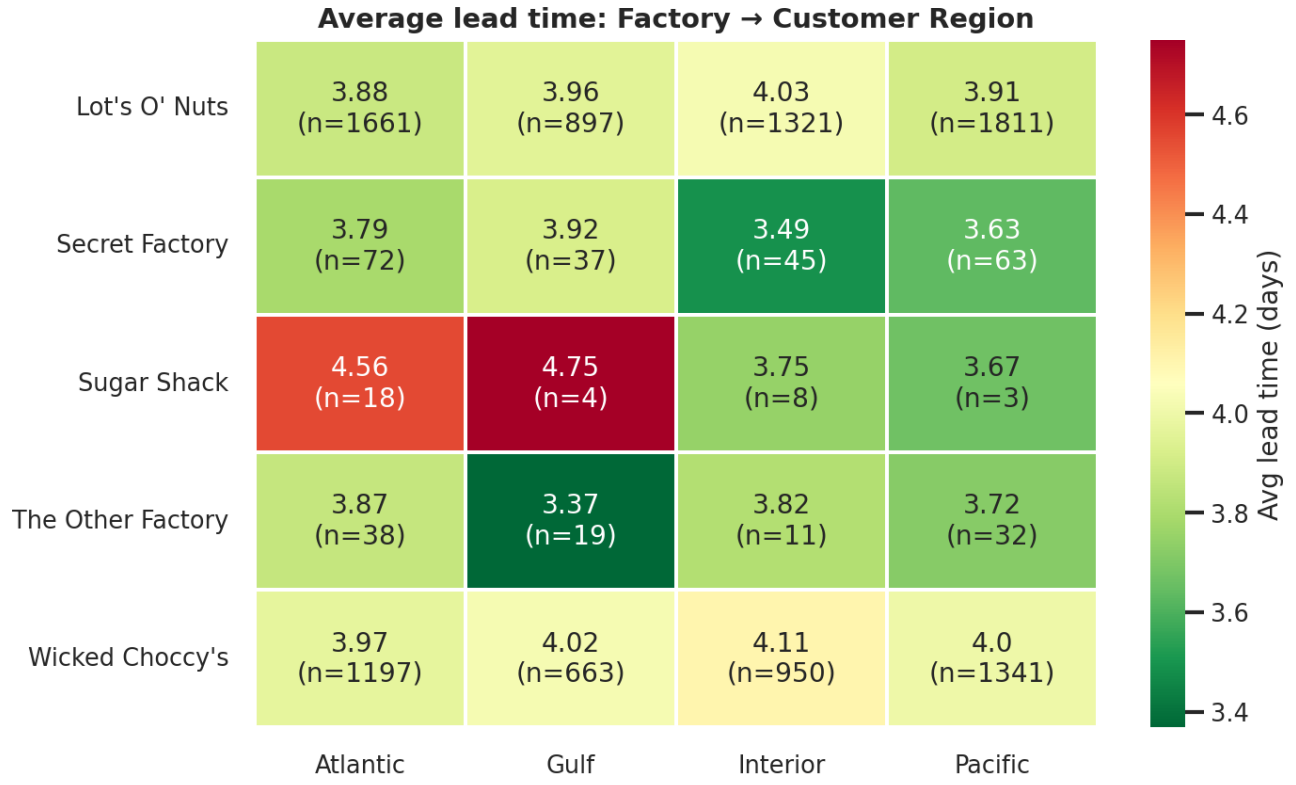


Figure 5. Average lead time by factory and customer region; sample sizes shown in each cell, and small cells are unreliable.

5.3 Geographic bottlenecks

Table 5. States flagged as bottlenecks (above-median volume and above-average lead time).

State	Shipments	Avg lead time	Delayed	Vs. national avg (days)
New York	1128	4.06	5.9%	+0.09
Illinois	492	4.10	4.9%	+0.13
Michigan	255	4.10	9.4%	+0.14
North Carolina	249	3.98	6.4%	+0.01
Arizona	224	4.05	11.2%	+0.09
Tennessee	183	4.37	9.8%	+0.40
Indiana	149	4.30	7.4%	+0.34
New Jersey	130	4.42	10.0%	+0.46
Oregon	124	4.35	4.0%	+0.39

Nine of the 40 states with at least 30 shipments meet the bottleneck definition. The largest by volume is New York (1,128 shipments) at only +0.09 days above average. No flagged state is more than about half a day above the national average (3.96 days). Taken with the permutation result, these differences are too small to call structural. The slowest states overall (New Mexico, Minnesota, Oklahoma, around 4.5 days) are also low-volume.

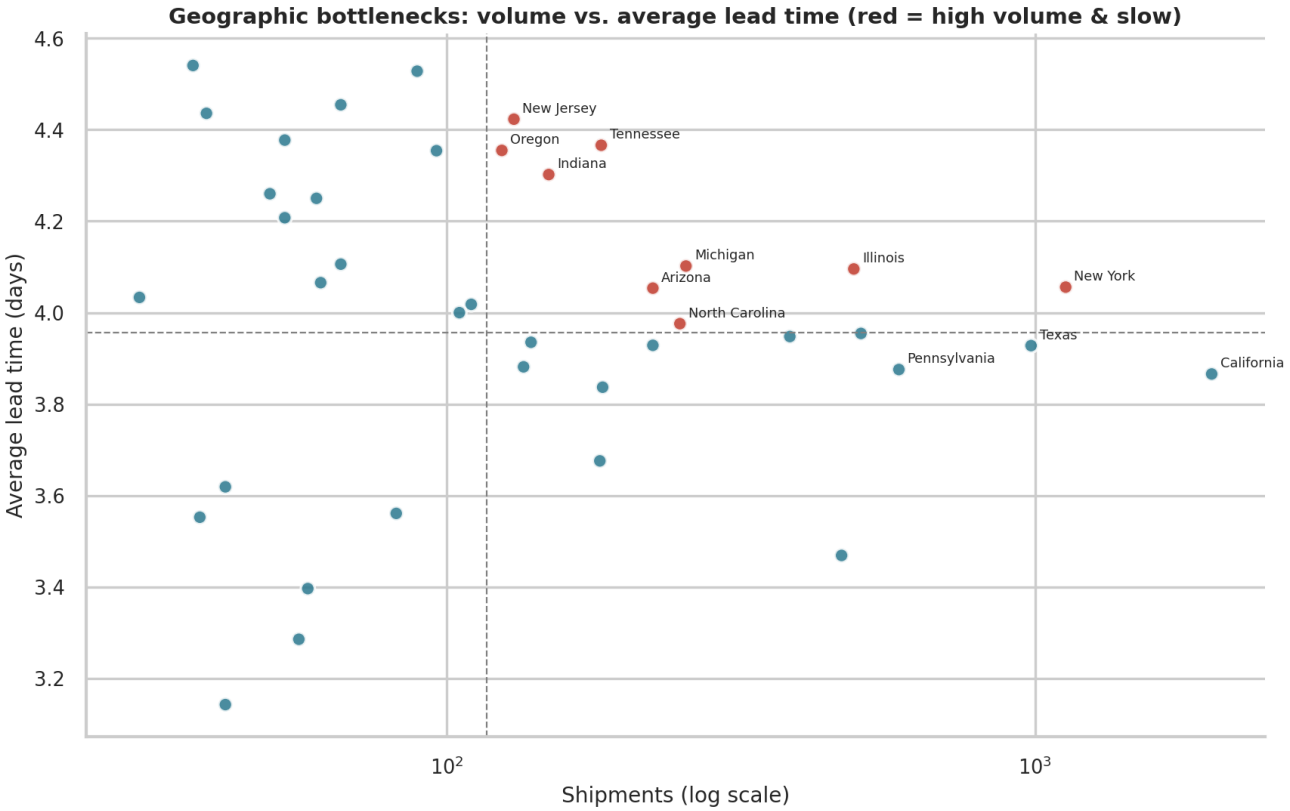


Figure 6. Volume versus average lead time by state; red marks high-volume, slower-than-average states.

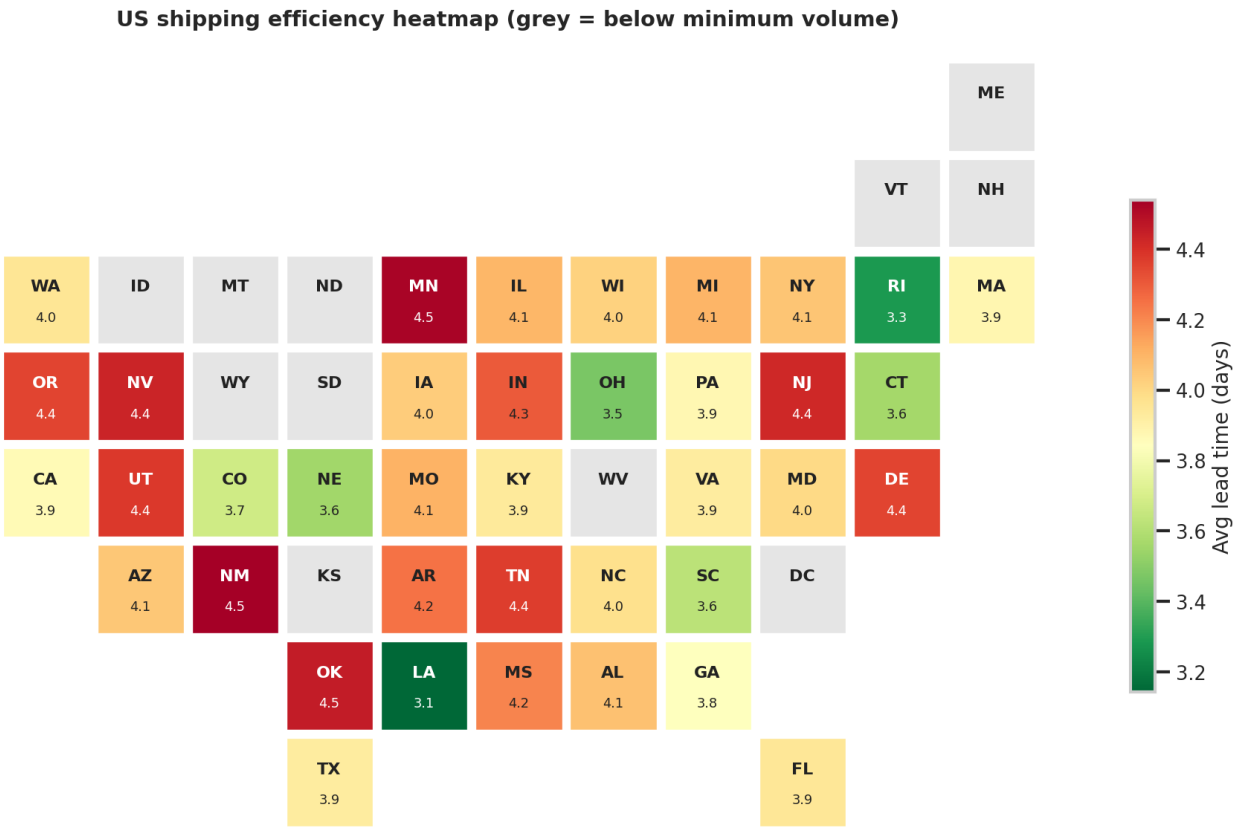


Figure 7. US tile-grid heatmap of average lead time; grey tiles have fewer than 30 shipments.

5.4 Statistical tests

Table 6. Kruskal-Wallis tests of lead time by grouping variable.

Grouping	H	p-value	Effect size (ε²)	Reading
Ship mode	6,052.9	<1e-300	0.594	Large
Customer region	9.9	0.019	0.001	Negligible
Factory	15.8	0.003	0.001	Negligible

Ship mode accounts for about 59% of the variation in lead-time ranks. Region and factory differences are statistically detectable only because the sample is large; each explains about 0.1% of the variation. Lead time is also essentially unrelated to order value and units (Spearman ρ ≈ 0.02 for both).

5.5 Sensitivity of delay frequency to the threshold

Table 7. Share of shipments classed as delayed at different thresholds.

Delay if lead time exceeds	Delayed shipments	All shipments	Standard Class only
4 days	4,075	40.0%	59.4%
5 days	1,864	18.3%	30.3%
6 days	648	6.4%	10.6%
7 days	24	0.2%	0.4%
8 days	2	0.0%	0.0%

The delay KPI is highly sensitive to its definition: a seven-day threshold flags almost nothing (0.2%), while a five-day threshold flags 18%. The six-day default is a judgement call that flags the slowest tail of Standard Class; the company's own service-level promise should replace it.

6. Discussion

- **Ship mode, not geography, drives lead time.** Differences between modes are measured in days; differences between regions, factories and states are measured in fractions of a day.

- **The route league table is not a reliable guide.** Average lead time per route is noisy (per-shipment SD ≈ 1.8 days with 30–260 shipments per route), and the observed route gap is within what random assignment produces. Reporting extremes without this check would overstate real differences.
- **The cost-time trade-off cannot be measured yet.** Gross margin is flat across modes and excludes freight, so the cost of faster shipping is invisible in these data.
- **Dispatch lead time is not delivery time.** Because only the ship date is recorded, the analysis measures how quickly orders leave the factory, not what customers experience.

7. Recommendations

1. **Fix the data at source.** Correct the ship-date export, and add actual delivery date and carrier/freight cost per shipment. This is the single biggest enabler of any further route optimisation.
2. **Manage the Standard Class tail.** Standard Class holds all delayed shipments. Review processing and dispatch for orders exceeding six days, and set an explicit service-level target to replace the placeholder threshold.
3. **Monitor routes with control limits, not rankings.** Track delay frequency and the efficiency score monthly for high-volume routes (for example Lot's O' Nuts and Wicked Choccy's to New York, Illinois, Michigan), and alert only when a route moves beyond normal variation.
4. **Do not re-route on current evidence.** The 1.4-day best-to-worst route gap is consistent with chance. Revisit factory assignment only after delivery-based lead times are available.
5. **Pilot targeted upgrades.** Where margin allows, trial Second Class on selected high-volume Standard Class lanes and measure the effect before wider rollout.

8. Limitations

- The date correction assumes Same Day shipments have a true lead time of zero; absolute lead-time levels depend on it.
- Lead time is order-to-ship, not order-to-delivery, and no freight cost is available.
- Factory-to-state coordinates could not be converted to distances because customer coordinates are absent; product-to-factory assignment is fixed, so factory effects cannot be separated from product effects.
- Small-volume factories (Sugar Shack, The Other Factory, Secret Factory) have too few shipments for reliable route comparison.
- 313 lines identical except for Row ID were retained; removing them would change results only marginally.

9. Conclusion

After correcting a systematic date offset, Nassau Candy's shipments leave the factory in about 4.0 days on average, with ship mode accounting for nearly all meaningful variation. Regional and route differences are small and statistically indistinguishable from random variation once sampling noise is considered. The practical conclusions are to manage the Standard Class tail, to avoid acting on route rankings built on thin samples, and to capture delivery dates and freight cost so the route-level optimisation the business wants can be done on solid evidence.

Appendix A. Reproducibility

All numbers, tables and figures in this paper are produced by a single Python script (pandas, NumPy, SciPy, Matplotlib, Seaborn). Run: `python nassau_candy_analysis.py --data Nassau_Candy_Distributor.csv --output outputs`. Options include `--delay-threshold`, `--min-shipments`, `--drop-duplicates` and `--no-calibrate`. The permutation test uses a fixed random seed (42).